Ashwini Kumar

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PROFESSIONAL SUMMARY

BCA Candidate (2026) and Full-Stack Software Developer specializing in Next.js, Node.js, Python, and Multi-Engine AI Automation. Track record of engineering low-latency systems, scalable asynchronous API workflows, and production-grade Web/AI solutions. Experienced in full-stack feature delivery, graph analytics, and automated digital infrastructure.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, TypeScript, C, C++, Java, PHP, SQL, HTML5, CSS3
- **Frameworks & Web Technology:** React.js, Next.js, Node.js, Express.js, Vite, jQuery, Bootstrap, RESTful APIs
- **AI & Data Engineering:** Generative AI & LLMs (Gemini API, Groq, Llama-3), Prompt Engineering, Nilearn, NetworkX, Pandas, Power BI, Edge-TTS, OpenCV, MLOps
- **Database & Developer Tools:** PostgreSQL, Git, GitHub, FFmpeg, Asyncio, SMTP Protocols, Linux CLI, Postman
- **Core Concepts:** Data Structures & Algorithms, Automated Pipelines, Cryptography, Asynchronous Processing, Systems Engineering

WORK EXPERIENCE

Software Developer Intern Sept 2025 – Oct 2025
Flocci *Ranchi, India*

- Engineered high-performance, server-side rendered (SSR) user interfaces using **Next.js** and **React.js**, reducing initial page load latency.
- Developed modular backend REST APIs in **Node.js** and **Express.js** to handle client-server data synchronization and user interactions.
- Collaborated in an agile full-stack environment to debug microservices, refine database queries, and integrate secure frontend components.

Social Media Analyst Intern Oct 2024 – Nov 2024
Mrs. India Inc *Jaipur, India*

- Managed data-driven performance analytics across digital channels, interpreting engagement metrics to refine social media growth strategy.
- Spearheaded post-production video editing and visual branding campaigns, resulting in measurable increases in reach and brand visibility.

KEY TECHNICAL PROJECTS

Dual-Engine AI Lead Extraction & Email Automation Bot *Python, Gemini API, Groq, SMTP, Pandas*

- Designed an asynchronous lead generation bot using a fallback multi-engine system (**Gemini 2.0 Flash** and **Llama 3.3 70B via Groq**) alongside RegEx parsing.
- Built automated batch-processing algorithms to extract unstructured contact leads, deduplicate data records, and log campaign histories.
- Integrated automated **SMTP MIME email pipelines** with dynamic field injections and automated PDF attachment handlers.

Autonomous AI Video Generation Bot *Python, Telegram API, Edge-TTS, FFmpeg, Pillow, MoviePy*

- Built an autonomous, event-driven video assembly engine utilizing **Python Asyncio** and **Telegram Bot API** for scheduled and on-demand content creation.
- Integrated **Gemini 2.5 Flash** to generate structured JSON scripts, dynamically mapping scene timings to **PEXELS API** stock video queries.
- Engineered a non-blocking media pipeline with **FFmpeg** overlay rendering, **Edge-TTS** audio synchronization, automated Pillow captions, and background garbage collection.

Synapse – Multilingual AI Conversation Platform *Python, Gemini API, Vite, JavaScript, CSS3*

- Developed a full-stack real-time AI conversation and translation engine powered by a **Python** API service layer and a lightweight **Vite** frontend.
- Implemented prompt engineering strategies to deliver precise cross-lingual translations and low-latency dynamic chat responses.

Comparative Analysis of Brain Connectivity (fMRI Graph Theory) *Python, Nilearn, NetworkX, fMRIPrep*

- Developed an fMRI signal processing pipeline to evaluate cortical network stability across motor lateralizations using graph theoretical metrics.
- Executed FDR-corrected statistical analysis on brain connectivity networks; research accepted/under publication for the Scopus-indexed **ICACSET** conference.

CHARR – Hybrid Encryption Framework for MMO Games *Python, Cryptography, RSA-2048, Algorithms*

- Architected a multi-layer hybrid encryption algorithm combining classical ciphers with RSA-2048 for real-time multiplayer chat security.
- Achieved sub-0.2s latency and 7.47 msg/sec throughput benchmarked on the DOTA-2 chat dataset; under publication for the Scopus-indexed **ICACSET** conference.

EDUCATION

Bachelor of Computer Applications (BCA) Expected: July 2026
Amity University Jharkhand, Ranchi, India

CERTIFICATIONS

- **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate** (Oct 2025)
- **Oracle Cloud Infrastructure 2025 Certified Data Science Professional** (Oct 2025)
- **The Complete Full-Stack Web Development Bootcamp** – Udemy (Jun 2025)
- **Career Essentials in Data Analysis** – Microsoft & LinkedIn (May 2025)
- **Building Systems with the ChatGPT API** – DeepLearning.AI (May 2025)

LANGUAGES

English (Professional Working), Hindi (Native/Bilingual), German (Basic)